

Spin-Density Matrix and Fragmentation Cross Section

1 Hard spin density matrix and Fragmentation Cross Section

In a common Cartesian spin basis,

$$\rho = \frac{1}{4} \left[\mathbb{I} \otimes \mathbb{I} + B_i \sigma^i \otimes \mathbb{I} + \bar{B}_j \mathbb{I} \otimes \sigma^j + C_{ij} \sigma^i \otimes \sigma^j \right], \quad i, j = x, y, z.$$

Thus B_i , $\bar{B}_j$, and C_{ij} are respectively the quark polarization, antiquark polarization, and spin-correlation tensor. Insert the two fragmentation spin factors directly on the two spin lines:

$$\begin{aligned} \mathcal{K} &= \text{Tr}_{s\bar{s}} \left\{ \rho (D_1 \mathbb{I} + a_i \sigma^i) \otimes (\bar{D}_1 \mathbb{I} + \bar{a}_j \sigma^j) \right\} \\ &= D_1 \bar{D}_1 + B_i \bar{D}_1 a_i + \bar{B}_j D_1 \bar{a}_j + C_{ij} a_i \bar{a}_j. \end{aligned}$$

Here $\text{Tr} \sigma^i = 0$ and $\text{Tr}(\sigma^i \sigma^j) = 2\delta^{ij}$. Equivalently,

$$\mathcal{K} = w_0 + B_i w_{B_i} + \bar{B}_j w_{\bar{B}_j} + C_{ij} w_{C_{ij}}, \quad (w_0, w_{B_i}, w_{\bar{B}_j}, w_{C_{ij}}) = (D_1 \bar{D}_1, \bar{D}_1 a_i, D_1 \bar{a}_j, a_i \bar{a}_j).$$

2 Calculate D_1 and a_i with the spin projector

The fragmentation correlator has open Dirac indices:

$$\Delta_{ab}(k; P_1, P_2) = \sum_X \int \frac{d^4 \xi}{(2\pi)^4} e^{ik \cdot \xi} \langle 0 | \mathcal{W} \psi_a(\xi) | P_1 P_2, X \rangle \langle P_1 P_2, X | \bar{\psi}_b(0) \mathcal{W} | 0 \rangle,$$

evaluated with $k^+ \equiv Q$ fixed (the frame with $P_{h\perp} = 0$, Q the large light-cone momentum of the fragmenting quark) and integrated over the small component k^- . At leading twist Δ is built entirely from the structures γ^- , $\gamma^- \gamma_5$, $i\sigma^{i-} \gamma_5$ ($i = x, y$). D_1 and a_i are the coefficients of these structures, read off by contracting Δ with the *conjugate* light-cone direction n ($n^2 = 0$, $n \cdot k = 1$, $\not{n} = \gamma^+ / Q$): since $\text{Tr}[\gamma^+ \gamma^-] \neq 0$ while $\text{Tr}[\gamma^- \gamma^-] = 0$, only γ^+ -type insertions have a nonzero overlap with Δ . Formally, write

$$\mathcal{D}(S) = \frac{1}{4z} \int dk^- \text{Tr}_D [\Delta \gamma^+ (1 + \gamma_5 \not{S})] \Big|_{\text{transverse } S} = D_1 + S_r a_r, \quad r = x, y,$$

which, using the exact identity $\gamma^+ \gamma_5 \not{\epsilon}_r = i\sigma^{r+} \gamma_5$ ($\not{\epsilon}_r = -\gamma^r$), gives

$$D_1 = \frac{1}{4z} \int dk^- \text{Tr}_D [\gamma^+ \Delta], \quad a_r = \frac{1}{4z} \int dk^- \text{Tr}_D [i\sigma^{r+} \gamma_5 \Delta], \quad r = x, y.$$

The longitudinal component is probed directly by γ_5 alone, without an extra slashed unit vector (inserting $\gamma_5 \not{\epsilon}_z$ would produce a structure orthogonal to every term in Δ):

$$a_z = \frac{1}{4z} \int dk^- \text{Tr}_D [\gamma^+ \gamma_5 \Delta].$$

Evaluating these three traces against the leading-twist parametrization of Δ gives

$$\begin{aligned} D_1 &= \frac{1}{4z} \int dk^- \text{Tr}_D [\gamma^+ \Delta], \\ a_z &= \frac{1}{4z} \int dk^- \text{Tr}_D [\gamma^+ \gamma_5 \Delta] = \frac{\epsilon_T^{mn} R_{Tm} k_{Tn}}{M_1 M_2} G_1^\perp, \\ a_r &= \frac{1}{4z} \int dk^- \text{Tr}_D [i\sigma^{r+} \gamma_5 \Delta] = \frac{\epsilon_T^{rn} R_{Tn}}{M_1 + M_2} H_1^\triangleleft + \frac{\epsilon_T^{rn} k_{Tn}}{M_1 + M_2} H_1^\perp, \quad r = x, y. \end{aligned}$$

For the reduced choice $H_1^\perp = 0$, with $\mathbf{k}_T = -\mathbf{P}_{hT}/z$ and $\epsilon_T^{xy} = 1$, abbreviate

$$\kappa = \frac{R_T}{M_1 + M_2} H_1^\triangleleft, \quad \lambda = \frac{R_T P_{hT}}{z M_1 M_2} G_1^\perp,$$

so that

$$\mathbf{a} \equiv \begin{pmatrix} a_x \\ a_y \\ a_z \end{pmatrix} = \begin{pmatrix} \kappa \sin \phi_R \\ -\kappa \cos \phi_R \\ \lambda \sin(\phi_R - \phi_h) \end{pmatrix}.$$

Antiquark side. The antiquark moves in the *opposite* direction, with large light-cone component $k'^- \equiv \bar{Q}$ instead of k'^+ . Its correlator $\bar{\Delta}$ is therefore built at leading twist from the *mirror* structures $\gamma^+, \gamma^+ \gamma_5, i\sigma^{i+} \gamma_5$, and $\bar{D}_1, \bar{a}_i$ must be extracted with the conjugate projectors $\gamma^-, \gamma^- \gamma_5, i\sigma^{r-} \gamma_5$ — every $+$ $\leftrightarrow$ $-$ exchanged relative to the quark case:

$$\bar{D}_1 = \frac{1}{4\bar{z}} \int dk'^+ \text{Tr}_D[\gamma^- \bar{\Delta}], \quad \bar{a}_z = \frac{1}{4\bar{z}} \int dk'^+ \text{Tr}_D[\gamma^- \gamma_5 \bar{\Delta}], \quad \bar{a}_r = \frac{1}{4\bar{z}} \int dk'^+ \text{Tr}_D[i\sigma^{r-} \gamma_5 \bar{\Delta}].$$

Charge conjugation relates the antiquark correlator to the quark one with an extra relative minus sign on every spin-dependent ($G_1^\perp$ - and H_1 -type) structure, while the spin-independent D_1 term is unaffected. Hence barring every fragmentation variable is *not* sufficient:

$$\bar{D}_1 = D_1|_{\text{barred}}, \quad \bar{a}_i = -a_i|_{\text{barred}}, \quad i = x, y, z,$$

where "barred" means every kinematic variable and fragmentation function on the right ($z, M_1, M_2, R_T, P_{hT}, \phi_R, \phi_h, H_1^\perp, G_1^\perp$) is replaced by its antiquark counterpart. With $\bar{\kappa}, \bar{\lambda}$ denoting κ, λ barred, this gives

$$\bar{\mathbf{a}} \equiv \begin{pmatrix} \bar{a}_x \\ \bar{a}_y \\ \bar{a}_z \end{pmatrix} = \begin{pmatrix} -\bar{\kappa} \sin \bar{\phi}_R \\ \bar{\kappa} \cos \bar{\phi}_R \\ -\bar{\lambda} \sin(\bar{\phi}_R - \bar{\phi}_h) \end{pmatrix}.$$

If only one fragmentation is observed, the spin factor reduces to $\mathcal{K}_{\text{one}} = D_1 + B_i a_i$.

3 Spin weights in terms of the fragmentation functions

With $\mathbf{a}, \bar{\mathbf{a}}$ as defined in Sec. 2, insert the results of Sec. 2 (with $H_1^\perp = 0$) into $\mathbf{w}_B = \bar{D}_1 \mathbf{a}$, $\mathbf{w}_{\bar{B}} = D_1 \bar{\mathbf{a}}$, and $W_C = \mathbf{a} \bar{\mathbf{a}}^T$. The single-spin weight vectors are

$$\mathbf{w}_B = \bar{D}_1 \begin{pmatrix} \kappa \sin \phi_R \\ -\kappa \cos \phi_R \\ \lambda \sin(\phi_R - \phi_h) \end{pmatrix}, \quad \mathbf{w}_{\bar{B}} = D_1 \begin{pmatrix} -\bar{\kappa} \sin \bar{\phi}_R \\ \bar{\kappa} \cos \bar{\phi}_R \\ -\bar{\lambda} \sin(\bar{\phi}_R - \bar{\phi}_h) \end{pmatrix},$$

and the correlation-weight matrix is the outer product

$$W_C = \mathbf{a} \bar{\mathbf{a}}^T = \begin{pmatrix} -\kappa \bar{\kappa} \sin \phi_R \sin \bar{\phi}_R & \kappa \bar{\kappa} \sin \phi_R \cos \bar{\phi}_R & -\kappa \bar{\lambda} \sin \phi_R \sin(\bar{\phi}_R - \bar{\phi}_h) \\ \kappa \bar{\kappa} \cos \phi_R \sin \bar{\phi}_R & -\kappa \bar{\kappa} \cos \phi_R \cos \bar{\phi}_R & \kappa \bar{\lambda} \cos \phi_R \sin(\bar{\phi}_R - \bar{\phi}_h) \\ -\lambda \bar{\kappa} \sin(\phi_R - \phi_h) \sin \bar{\phi}_R & \lambda \bar{\kappa} \sin(\phi_R - \phi_h) \cos \bar{\phi}_R & -\lambda \bar{\lambda} \sin(\phi_R - \phi_h) \sin(\bar{\phi}_R - \bar{\phi}_h) \end{pmatrix}.$$

Every entry of $\mathbf{w}_B$, $\mathbf{w}_{\bar{B}}$, and W_C is thus expressed directly in terms of the collinear-times-TMD fragmentation functions D_1 , $H_1^\perp$, $G_1^\perp$ (and their barred counterparts) together with the observed hadron-pair kinematics $R_T, P_{hT}, \phi_R, \phi_h$ (and barred).

4 Deriving $\mathcal{K}$ with the quark mass kept

Specialize Sec. 1 to $e^+ e^- \rightarrow \gamma^* \rightarrow q \bar{q}$, with θ the quark production angle and

$$\rho_m \equiv \frac{4m^2}{s}, \quad A_0 = 1 + \cos^2 \theta, \quad B_0 = \sin^2 \theta, \quad A_U = A_0 + \rho_m B_0.$$

Mass can enter through the hard vertex ($B_i, \bar{B}_j, C_{ij}$) or the fragmentation vertex (D_1, a_i).

Contracting the leptonic tensor with the covariant spin sums $u(k, S) \bar{u}(k, S) = \frac{1}{2}(\not{k} + m)(1 + \gamma_5 \not{\epsilon})$, $v(k', S') \bar{v}(k', S') = \frac{1}{2}(\not{k}' - m)(1 + \gamma_5 \not{\epsilon}')$ at the photon vertex gives the standard exact-in- m result: parity still forces $B_i = \bar{B}_j = 0$, and in the $(x, y, z) = (-r, n, k)$ axes of Sec. 1 (k along the quark momentum, r in the lepton-quark plane, n normal to it),

$$C = \frac{1}{A_U} \begin{pmatrix} (1 + \rho_m) B_0 & 0 & -2\sqrt{\rho_m} \sin \theta \cos \theta \\ 0 & -(1 - \rho_m) B_0 & 0 \\ -2\sqrt{\rho_m} \sin \theta \cos \theta & 0 & A_0 - \rho_m B_0 \end{pmatrix}.$$

Mass terms from the final-state quark propagator

Decompose the exact final-state spin sum by light-cone power,

$$\not{k} + m = k^+ \gamma^- + (\not{k}_\perp + m) + k^- \gamma^+, \quad k^- = \frac{m^2 + \mathbf{k}_\perp^2}{2k^+},$$

and dress Δ 's known structures with the full propagator instead of its truncation $k^+ \gamma^-$:

$$\Delta_{\text{massive}} = (\not{k} + m) D_1 + \gamma_5 \gamma^0 \gamma^3 (\not{k} + m) X_z G_1^\perp + \gamma^r \gamma_5 (\not{k} + m) Y_r,$$

with X_z, Y_r as in Sec. 2. Every trace of the subleading $(\not{k}_\perp + m)$ and $k^- \gamma^+$ blocks against the Sec. 2 projectors $\gamma^+, \gamma^+ \gamma_5, i\sigma^{r+} \gamma_5$ vanishes identically (verified by explicit Dirac-trace computation, keeping $m, k^-, \mathbf{k}_\perp$ fully general):

$$\text{Tr}_D[\gamma^+(\not{k}_\perp + m)] = \text{Tr}_D[\gamma^+ \gamma^+] = \text{Tr}_D[\gamma^+ \gamma^0 \gamma^3 (\not{k}_\perp + m)] = \text{Tr}_D[\gamma^+ \gamma^0 \gamma^3 \gamma^+] = \text{Tr}_D[i\sigma^{r+} \gamma_5 (\not{k}_\perp + m)] = \text{Tr}_D[i\sigma^{r+} \gamma_5 \gamma^+] = 0.$$

So D_1, a_z, a_r are unchanged at any order in m or $\mathbf{k}_\perp$: the quark mass cannot enter through these three Dirac channels at all.

It can only enter through a channel Sec. 2 never uses: the identity. The scalar term in $\not{k}_\perp + m$ generates $\Delta \supset \mathbb{I} \cdot E(z, \mathbf{k}_\perp; m)$, read off by the scalar trace $E = \frac{1}{4z} \int dk^- \text{Tr}_D[\Delta]$. Since $\text{Tr}_D[\not{k} + m] = 0 + 4m$, this ansatz gives $E = mD_1$ directly from the $+m$ in the propagator. But $\mathcal{K} = \text{Tr}_{s\bar{s}}[\rho(D_1 \mathbb{I} + a_i \sigma^i) \otimes (\bar{D}_1 \mathbb{I} + \bar{a}_j \sigma^j)]$ traces over quark-spin space, not Dirac space, so E cannot correct D_1 or a_i inside it — it would enter only through a separate $E\bar{D}_1 + D_1\bar{E}$ term alongside $\mathcal{K}$, left out below.

Result

$D_1, a_i, \bar{D}_1, \bar{a}_i$ are mass-independent, so all of $\mathcal{K}$'s mass dependence sits in C . With $B_i = \bar{B}_j = 0$, $\mathcal{K} = D_1 \bar{D}_1 + C_{ij} a_i \bar{a}_j$ is the bilinear form

$$\mathcal{K} = D_1 \bar{D}_1 + \mathbf{a}^T C \bar{\mathbf{a}},$$

using the same $\mathbf{a}, \bar{\mathbf{a}}$ of Sec. 2. Expanding the product gives

$$\begin{aligned} \mathcal{K} = D_1 \bar{D}_1 &+ \frac{1}{A_U} \left[B_0 \kappa \bar{\kappa} \cos(\phi_R + \bar{\phi}_R) - A_0 \lambda \bar{\lambda} \sin(\phi_R - \phi_h) \sin(\bar{\phi}_R - \bar{\phi}_h) \right] \\ &+ \frac{\sqrt{\rho_m} \sin 2\theta}{A_U} \left[\kappa \bar{\lambda} \sin \phi_R \sin(\bar{\phi}_R - \bar{\phi}_h) + \lambda \bar{\kappa} \sin(\phi_R - \phi_h) \sin \bar{\phi}_R \right] \\ &+ \frac{\rho_m B_0}{A_U} \left[\lambda \bar{\lambda} \sin(\phi_R - \phi_h) \sin(\bar{\phi}_R - \bar{\phi}_h) - \kappa \bar{\kappa} \cos(\phi_R - \bar{\phi}_R) \right], \end{aligned}$$

whose three lines are the massless Collins-type correlation, an $O(\sqrt{\rho_m})$ term mixing the transverse sector of one jet with the longitudinal sector of the other (vanishing at $m = 0$), and an $O(\rho_m)$ correction with the same sum-to-difference angle flip found in the exact massive double-transverse and double-Collins $e^+ e^-$ cross sections.